Arav Palsule

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LinkedIn — GitHub

B.E. Data Science (2027) — CGPA: 8.34/10

Professional Summary

Aspiring Data Analyst with a strong foundation in statistics, data analysis, and quantitative problem-solving. Skilled in Python, SQL, data cleaning, exploratory data analysis (EDA), data visualization, and statistical analysis. Experienced in working with historical datasets, performing feature engineering, analyzing trends, building forecasting models, and evaluating results using statistical metrics. Interested in transforming raw data into actionable insights and supporting data-driven decision-making.

Education

A. P. Shah Institute of Technology, Thane 2023 – 2027
Bachelor of Engineering in Data Science **CGPA: 8.34/10**

Technical Skills

Programming: Python, SQL
Data Analysis: Data Cleaning, Data Wrangling, Exploratory Data Analysis (EDA), Data Preprocessing, Feature Engineering, Trend Analysis
Statistical Analysis: Descriptive Statistics, Statistical Analysis, Correlation Analysis, Regression Analysis, Hypothesis Testing
Python Libraries: Pandas, NumPy, Matplotlib, Seaborn
Databases: MySQL, SQLite, SQL Queries, Data Retrieval
Data Science: Predictive Analytics, Forecasting, Model Evaluation, MAE, RMSE, SMAPE
Tools and Platforms: Git, GitHub, Flask, Supabase, Excel

Projects

Dengue Outbreak Prediction – Data Analysis and Forecasting
Analyzed historical dengue case reports alongside weather variables, geographic attributes, and temporal features to identify patterns relevant to outbreak forecasting. Performed data preprocessing, feature engineering, lag analysis, and rolling-statistic calculations to prepare structured datasets for regression modelling. Trained and compared Random Forest, Extra Trees, HistGradientBoosting, and Ridge regression models using log-transformed target values. Evaluated forecasting performance through chronological holdout and rolling-window validation using MAE, RMSE, and SMAPE.

SPHERE AI – Educational Data Analytics Platform
Developed a Flask-based educational analytics platform with dashboards for monitoring student performance and engagement. Performed Exploratory Data Analysis (EDA) on assessment and engagement datasets to identify learning patterns and performance trends. Designed data processing workflows and academic KPI reporting to support data-driven educational insights.

The Review Room – Review and Text Analytics
Worked with a large-scale review dataset to analyze customer feedback and extract meaningful patterns from textual data. Implemented text processing, sentiment analysis, topic exploration, and review retrieval workflows to organize and analyze user feedback. Explored review trends and aspect-level insights to support structured interpretation of customer opinions.

CrackX – Road Condition Data Management
Developed a road condition monitoring platform to organize and process road damage reports and associated image and video data. Integrated YOLO-based detection outputs, including damage categories, bounding boxes, and severity information, into road condition reporting workflows.

Experience

AI Intern – Ennoble Social Innovation
September 2026 – October 2026

- Created custom Claude Skills using context engineering to automate business tasks, including PDF formatting according to internal document structures and standards.
- Developed an employee focus-monitoring prototype integrating face detection and emotion analysis to estimate focus scores and identify potential sleep or inactivity events.
- Designed automated screenshot capture and HR reporting workflows to communicate detected events and employee focus patterns.
- Built workflow automations using Zapier, LLMs, and API integrations to streamline repetitive operational processes.

Achievements

- **Rank 10** – C.R. Rao Statistics & Probability Olympiad: Ranked among 2.5 lakh+ participants across Asia.
- **AIR 3192, AICR 35** – Aryabhatta National Maths Competition 2026: Scored 100% in Round 1 and 80% in Round 2.
- **2nd Runner-Up** – QUASAR 4.0 Hackathon: Delivered a technical solution under a 24-hour development deadline.
- **Elite Rank** – NPTEL Multivariable Calculus: Recognized for mathematical and quantitative problem-solving skills.

Leadership

Founding President – Maths & Applied Science Club — 2026

- Founded and led the club, organizing 25+ technical events to promote mathematics, science, and technical learning among students.

Research Member – DSA Club — 2025 – 2026

- Explored data science, machine learning, and analytical workflows; mentored junior members in technical concepts.